

Project Report Audit

Audited against [C:/Users/ayush/Desktop/VIT_BTech_Report_Ayush.pdf](#).

Coverage Matrix

Report claim / requirement	Status	Evidence
5-year NASA POWER dataset with 52,608 hourly rows	Present	cleaned_nasa_power.csv has 52,608 rows from 2020-01-01 05:30:00 to 2026-01-01 04:30:00
LSTM solar and load forecasting	Present	solar model: True, load model: True, solar RMSE 3.5749, load RMSE 4.7597
PPO reinforcement learning dispatch agent	Present	model exists: True, timesteps: 20000
180 kWh BESS with physics constraints	Present	capacity 180.0 kWh, SoC limits 20.0-90.0%, charge 55.0 kW, discharge 58.0 kW
Time-of-use tariff arbitrage	Present	tariff values in EMS dataset: [2.6, 5.6, 9.2]
FastAPI backend endpoints	Present	/forecast, /optimize, /decisions, /metrics, /alerts, /simulate, /matlab/export, and live simulator endpoints are implemented
SCADA-style operator dashboard	Improved	React dashboard now has command bar, dense KPIs, live stream controls, override controls, charts, alerts, cost, sustainability, scenario and dispatch table
Explainable dispatch reasoning	Present	Dispatch table and recommendation panel expose action and reason strings from backend dispatch logic
Local fallback / dynamic simulator	Present	Frontend cache exists and backend live telemetry generator can be started from dashboard button
MATLAB/Simulink export	Present	CSV, MAT, validation reference CSV, and MATLAB runner script exist in data/exports
Detailed EDA	Present	Generated docs/data-eda-report.md and plots under data/outputs/eda

Gaps To Be Aware Of

- The report describes a continuous action scalar PPO formulation, while the current environment uses a discrete action space: idle, charge, discharge. This is acceptable for the original dashboard requirement but does not exactly match the continuous-control claim.
- The report mentions wind generation in the MDP state. The implementation uses wind speed as a data feature, but no wind turbine power model is dispatched.

- The current regenerated EDA finds strongest temperature-load correlation at 0 hours, while the report text claims an approximately 2.5-hour lag. The report should be updated or the load generator should be changed to include a clearer lagged HVAC response.
- The report text mentions 20%-95% SoC limits in some places. The implemented BESS uses 20%-90%, which is more conservative but should be made consistent in the report or code.
- A MATLAB/Simulink export workflow exists, but a complete .slx physical model is not present in the project folder.
- The dashboard can simulate live telemetry, but it is not connected to real SCADA, smart meters, inverter telemetry, or BMS hardware.

Novelty Assessment

- The strongest project novelty is not simply using LSTM or PPO, because both are known in energy management research.
- The useful differentiator is the operator-trust layer: dispatch reasoning, hard battery safety boundaries, override modes, fallback telemetry, scenario stress testing, and MATLAB export in one teaching-grade EMS stack.
- The new live simulator is valuable for review and demos because it creates a control-room experience without needing real SCADA access while still enforcing realistic battery and tariff behavior.
- A stronger publishable novelty would be: hybrid XAI + safety-shielded DRL EMS with operator override and digital-twin fallback telemetry for low-infrastructure microgrid validation.

Report Text Signals Used

The report emphasizes LSTM forecasting, PPO-based BESS dispatch, 52,608 NASA POWER rows, 180 kWh BESS, ToU arbitrage, SCADA-style explainability, operator override, fallback simulation, MATLAB/Simulink export, and 12.23% cost savings. The current project covers most of that system surface, with the main technical mismatch being discrete PPO action control versus the report's continuous-action wording.